VERY AMPLENESS FROM SEPARATING CLOSED POINTS AND NONZERO TANGENT VECTORS BY SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains two bounded asides, '(computation omitted)' at the injectivity on closed points and '(small detail omitted)' at the singleton fibre, both reproduced verbatim; the surrounding steps are otherwise complete as written.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-separate-points-tangent-vectors is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Canonical morphism to projective space attached to a subspace of global sections. Constructions, Example constructions-example-projective-space, invoked at proof lines 4328-4335. Supplies the morphism phi_{L,V}: X to P(V) whose closed-immersion property is at stake.
- Image of a proper scheme is closed and a proper morphism with finite fibres is finite. Morphisms, Lemma morphisms-lemma-image-proper-scheme-closed and Cohomology of Schemes, Lemma coherent-lemma-proper-finite-fibre-finite-in-neighbourhood, invoked at proof lines 4336-4342. Supplies properness of phi and finiteness once the fibres are singletons.
- Tangent space of a projective space at a rational point and the cotangent description of surjectivity. Same chapter, Lemma lemma-tangent-space-rational-point, invoked at proof lines 4374-4377. Supplies that injectivity of the tangent map is surjectivity of m_p/m_p^2 to m_x/m_x^2.
- Local criterion for surjectivity of a map of local rings with surjective cotangent map. Algebra, Lemma algebra-lemma-when-surjective-local, invoked at proof lines 4379-4381. Supplies the final surjectivity of the stalk map from the finite module structure and the surjection on cotangent spaces.

Source reference. Original lemma 0E8M: sources/stacks-chow/tags.txt, source lines 17822–17822. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02OH: sources/stacks-chow/tags.txt, source lines 3516–3516. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original definition 0FCY: sources/stacks-chow/tags.txt, source lines 19198–19198. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0B2E: sources/stacks-varieties/pinned-original-source.tex, source lines 3059–3059. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be an algebraically closed field. Let X be a proper k-scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $V \subset H^0(X, \mathcal{L})$ be a k-subvector space. If*

1. *for every pair of distinct closed points $x, y \in X$ there is a section $s \in V$ which vanishes at x but not at y, and*
2. *for every closed point $x \in X$ and nonzero tangent vector $\theta \in T_{X/k,x}$ there exists a section $s \in V$ which vanishes at x but whose pullback by θ is nonzero,*

then $\mathcal{L}$ is very ample and the canonical morphism $\varphi_{\mathcal{L},V} : X \to \mathbf{P}(V)$ is a closed immersion.

Proof. Condition (1) implies in particular that the elements of V generate $\mathcal{L}$ over X. Hence we get a canonical morphism

$$\varphi = \varphi_{\mathcal{L},V} : X \longrightarrow \mathbf{P}(V)$$

by Constructions, Example 0FCY. The morphism φ is proper by Morphisms, Lemma 01W6. By (1) the map φ is injective on closed points (computation omitted). In particular, the fibre over any closed point of $\mathbf{P}(V)$ is a singleton (small detail omitted). Thus we see that φ is finite, for example use Cohomology of Schemes, Lemma 02OH. To finish the proof it suffices to show that the map

$$\varphi^\sharp : \mathcal{O}_{\mathbf{P}(V)} \longrightarrow \varphi_* \mathcal{O}_X$$

is surjective. This we may check on stalks at closed points. Let $x \in X$ be a closed point with image the closed point $p = \varphi(x) \in \mathbf{P}(V)$. Since $\varphi^{-1}(\{p\}) = \{x\}$ by (1) and since φ is proper (hence closed), we see that $\varphi^{-1}(U)$ runs through a fundamental system of open neighbourhoods of x as U runs through a fundamental system of open neighbourhoods of p. We conclude that on stalks at p we obtain the map

$$\varphi_x^\sharp : \mathcal{O}_{\mathbf{P}(V),p} \longrightarrow \mathcal{O}_{X,x}$$

In particular, $\mathcal{O}_{X,x}$ is a finite $\mathcal{O}_{\mathbf{P}(V),p}$-module. Moreover, the residue fields of x and p are equal to k (as k is algebraically closed – use the Hilbert Nullstellensatz). Finally, condition (2) implies that the map

$$T_{X/k,x} \longrightarrow T_{\mathbf{P}(V)/k,p}$$

is injective since any nonzero θ in the kernel of this map couldn't possibly satisfy the conclusion of (2). In terms of the map of local rings above this means that

$$\mathfrak{m}_p/\mathfrak{m}_p^2 \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2$$

is surjective, see Lemma 0B2E. Now the proof is finished by applying Algebra, Lemma 0E8M. □